



CENTRAL CAMPUS PARKING
REDEVELOPMENT PROJECT: MULTI-LEVEL
PARKING STRUCTURE FOR THE SHIDLER–
GEORGE HALL–ARCHITECTURE LOT

UNIVERSITY OF HAWAI'I AT MANOA

ITM-321-001

COURSE CODE: 84253

PROJECT MANAGEMENT

NICHOLAS R. DUNCAN

MAY 14TH, 2026



TABLE OF CONTENTS

1. Introduction.....	3
2. Project Overview	3
Team Roles Overview.....	4
3. Needs Assessment / Feasibility Analysis.....	4
4. Issues and Opportunity Analysis.....	5
5. Market Research Analysis	6
6. Scope Analysis.....	7
In Scope	7
Out of Scope	8
High-Level Cost Summary.....	8
7. Tangible Deliverables	9
8. Requirements Analysis	9
Requirements Traceability	10
9. User Stories, Use Case, AND Swimlane Diagrams	11
User Stories	11
Use Case: Parking in the New Structure	11
Swimlane Diagrams	12
10. Risk Assessment and Risk Mitigation	13
Risk 1: Structural or Site Feasibility Risk.....	13
Risk 2: Budget Overrun Risk	13
Risk 3: Temporary Parking Loss During Construction.....	14
Risk 4: Traffic Flow Risk	14
Risk 5: Stakeholder Resistance.....	14
Risk 6: Technology Integration Risk	14
Risk 7: Native Population Cultural Aspects.....	14
11. Work Breakdown Structure	15
12. Timeline / Milestone Map.....	16
13. Next Steps	17
14. Conclusion	18
Appendices.....	19
Appendix A: Requirements Analysis	19



Appendix B: Requirements Traceability	20
Appendix C: Swimlane Diagrams	21
Appendix C1: Issue Management Workflow.....	21
Appendix C2: Risk Management Workflow.....	22
Appendix C3: Change Management Workflow.....	23
Appendix D: Work Breakdown Structure	24
Appendix E: Timeline.....	25
Appendix F: Cost Breakdown	26
Appendix G: Project Roles	27
References.....	28



1. INTRODUCTION

Parking is one of the most visible and recurring infrastructure problems on commuter-heavy university campuses. When parking supply is not aligned with demand by time and location, students, faculty, staff, and visitors lose time, experience frustration, and contribute to unnecessary congestion. This is especially true in campus core areas, where the value of close-in parking is much higher than parking located on the periphery. At the University of Hawai‘i at Mānoa, parking is not simply a convenience issue. It affects access to instruction, timeliness, circulation, and the overall user experience for commuters. UH Mānoa’s Campus Transportation Demand Management Plan documented 5,497 parking spaces campus-wide, with an adjusted commuter supply of 4,378 spaces and peak commuter plus faculty/staff demand reaching 3,939 spaces, or about 90 percent occupancy. The same document notes that upper campus lots were near 100 percent utilization during working hours. Those findings matter because parking systems become functionally difficult to use before they are mathematically full.

This paper proposes a project to redevelop the surface parking lot located directly south of Shidler College of Business, west of George Hall, and north of the Architecture Building into a multi-level parking structure with smart parking technology. The proposal is based on a straightforward idea: this is a high-value academic-core site currently being used as a relatively low-density surface lot. In project management terms, that creates a strong opportunity for improvement through better land utilization, measurable capacity gains, and improved commuter access. The project is also well-suited to the assignment because it allows for a full feasibility analysis, detailed requirements definition, and a predictive work breakdown structure.

2. PROJECT OVERVIEW

The proposed project is the redevelopment of the existing central-campus surface lot into a structured parking facility designed to serve commuter students, faculty, staff, and authorized visitors. The garage would include not only additional parking spaces but also a smart parking system consisting of occupancy sensors, digital entrance signage, permit integration, wayfinding, lighting, security features, and ADA-compliant access. The justification for this project is not that UH Mānoa has no parking. That claim would be too broad and not supported by the available numbers. A more accurate and defensible statement is that parking access in the central academic core remains constrained during peak periods, even though total campus supply exists elsewhere. The official campus plan already showed very high peak occupancy and near-full upper-campus utilization, and more recent commuter-service notices show



that student permits sold out in multiple semesters. That combination supports the argument that core-campus parking access remains a real operational issue.

The proposed facility would likely be a 4-level above-ground garage, sized to fit the current lot's footprint. Because there is no published engineering drawing in the materials available here, capacity should be presented as a planning estimate rather than an official university figure. A reasonable project estimate is that the existing lot holds roughly 70 to 100 vehicles, while a structured replacement could provide roughly 300 to 450 spaces, depending on circulation, ramps, setbacks, and structural design. On a more specific level, for this structure, the likelihood is 320 spaces. That would create a meaningful net gain in upper-campus capacity without requiring new land acquisition. A “done” project in this sense is a structurally adequate, fully outfitted 4-level parking structure with street-side signaling, occupancy detectors, and displays, containing at least 300 spaces.

TEAM ROLES OVERVIEW

Successful delivery of the UH Mānoa Central Campus Parking Redevelopment project requires clearly defined leadership roles distributed across the university, design consultants, construction management, regulatory bodies, and internal finance and procurement functions. The following section identifies the twelve principal head-level roles that carry decision-making authority throughout the project lifecycle, from initial feasibility approval through construction closeout and operational handoff. Each role is tied to a specific organizational owner and carries a distinct area of responsibility aligned to the project's Work Breakdown Structure and governance framework.

Role assignments span five functional clusters: University Leadership, which holds budget authority and defines operational requirements; Design Team Leads, who are responsible for architectural, structural, and civil design; Construction Leadership, which manages execution under a Construction Manager at Risk delivery model; Regulatory and Compliance contacts, who govern state and municipal permit approvals; and Finance and Procurement leads, who authorize capital funding and manage competitive bidding. Together, these roles form the decision-making backbone of the project and establish clear accountability at every stage. See [Appendix G](#) for more details.

3. NEEDS ASSESSMENT / FEASIBILITY ANALYSIS

The first part of the feasibility case is the scale of the university itself. UH Mānoa's 2023–2024 Common Data Set reports 19,256 total students, including 14,576 undergraduates and 4,680 graduate students. That enrollment level alone makes parking and circulation significant operational concerns,



especially on a campus where many users commute. The second part is the actual parking system performance documented by UH Mānoa. The Campus Transportation Demand Management Plan reported 5,497 total parking spaces campus-wide, with approximately 80 percent allocated to commuter parking, equal to 4,423 stalls, later adjusted to 4,378 after projected construction-related losses. Peak commuter plus faculty/staff demand was measured at 3,939 occupied spaces, which translates to roughly 90 percent occupancy of the adjusted supply. The plan states that upper campus lots were near 100 percent utilization during working hours. From a planning perspective, those are important numbers because parking systems become unreliable when occupancy approaches the mid-80 percent range and above. FHWA parking guidance treats about 85 percent occupancy as an efficient target, and smart parking technology literature notes that sensor-supported systems can support slightly higher operational occupancy by improving information reliability. Even at a low rate of all daily 12,000 estimated visitors commuting with cars at 60%, we still have a huge discrepancy.

The third part of the feasibility case is operational evidence from recent years. UH Mānoa Commuter Services announced that all individual student car permits were sold out for Spring 2022 and that all individual car permits were reserved out for Fall 2022. This has been the case and will be the case until the parking issue is resolved. Those announcements do not prove that every lot is full every hour, but they do show that parking access is constrained enough that the university rations parking through class-standing-based allocation and reservation. That supports the practical argument that the current system is under pressure from user demand. This project is especially feasible because the proposed site is already dedicated to parking. That reduces the complexity associated with land acquisition, rezoning, or conversion of an unrelated facility. In project terms, this means the proposal is an upgrade and intensification of existing parking use, not a totally new land use. That makes the concept more realistic, more defensible, and easier to scope.

4. ISSUES AND OPPORTUNITY ANALYSIS

For this project, the key issue is not simply “there is not enough parking.” The issue is more specific: high-value core-campus parking is scarce during the times people most need it, and the current surface lot uses valuable land inefficiently. The first user pain point is searching for time. When lots are near full, drivers spend more time circulating and less time moving directly to their destination. That creates frustration, delays, and localized congestion. Smart parking literature repeatedly identifies search inefficiency as one of the major reasons digital occupancy systems are useful. Construction of a 1,100-space structure by the Korean studies building is set to be developed, but this does not address the issue of



the commuters on the western side of campus. This cuts into the deficit of parking on campus, but it is not a campus-wide fix. Using the preexisting lot by Shidler College relieves parking issues at a much more important pain point and in a high-traffic area. Construction of this major structure without supporting lots around campus will just create congestion in the central campus areas during rush hours.

The second pain point is uncertainty. A commuter can leave home on time and still do not know whether parking near class will be available. That uncertainty affects planning, class punctuality, and overall perceived accessibility of the campus. When central lots or nearby structures are close to full, students often must circle, park farther away than intended, or arrive earlier than necessary just to secure a space. The third pain point is inefficient land use. A surface lot in the center of the academic core provides relatively few spaces for the land it consumes. In a central zone bordered by Shidler, George Hall, and Architecture, that land has unusually high transportation value. Structured parking exists specifically because vertical development allows much higher capacity on the same footprint. The opportunity, then, is not just to build more stalls. It is to create a more efficient, safer, and more predictable parking system in a location where proximity matters. This transforms a general complaint into a real project opportunity. The lot can be redeveloped into a structured facility that addresses capacity, access, safety, and information flow all at once.

5. MARKET RESEARCH ANALYSIS

In this context, market research is not about commercial competitors in the traditional sense. It is about identifying industry practices, transportation-planning norms, and comparable institutional logic that support the project concept. One important benchmark is the idea of target occupancy. Transportation and parking literature consistently points to around 85 percent occupancy as the threshold at which parking remains usable without excessive cruising. FHWA materials cite 85 percent as a target for efficient operation, while later FHWA guidance notes that in off-street facilities, better sensor data can support occupancy in the 90 to 92 percent range because drivers can be guided more accurately to available spaces. That matters because UH Mānoa's documented 90 percent occupancy already places the campus near the range where efficient management becomes difficult without better information systems. We see these occupancy rates at peak times, indicating a parking shortage. A second benchmark is the economics of structured parking. VTPI's 2025 parking cost and supply analysis explains that structured parking is substantially more expensive than surface parking but can become cost-effective when land values are high and space is limited. The report notes that structured parking typically costs far more per space than surface lots and becomes more rational where high-value land would otherwise be consumed



by low-density parking. That describes the central academic core well, especially in the area around Shidler: the lot is valuable because of where it is, not just what it currently does.

A third benchmark is the use of smart parking systems. Reviews in *Heliyon* and *Smart Cities* describe smart parking systems as tools that improve occupancy detection, user information, and parking search efficiency through sensors, networking, and interface systems. In simple terms, smart parking technology does not create spaces, but it helps users find available spaces faster and helps operators manage the facility better. For a university project, that means the best practice solution is not merely concrete and striking. It is a combination of structured parking and information systems. Taken together, market research supports the project concept. The literature suggests that parking systems should avoid sustained near-full occupancy, that central valuable land justifies structured parking more than remote land does, and that smart parking systems improve usability in high-demand environments.

6. SCOPE ANALYSIS

IN SCOPE

This project includes the full planning and delivery of a structured parking redevelopment on the selected site. In scope are:

- Project charter and approval package
- Business Needs Assessment
- Issues and opportunity assessment
- Market research and benchmarking
- Return on investment planning estimate
- Stakeholder analysis
- Budget analysis
- Milestone map and timeline
- Requirements elicitation and specification
- Site review, geotechnical, and circulation analysis



- Concept, architectural, structural, and systems design
- Demolition or clearing of the current surface lot as needed
- Construction of a 4- to 5-level above-ground parking structure
- Installation of lighting, wayfinding, security cameras, ADA elements, elevators, and stairs
- Installation of smart parking occupancy sensors and digital entry signage
- Permit and access integration with UH parking operations
- Testing, commissioning, training, handoff, and closeout documentation

OUT OF SCOPE

This project does **not** include a total redesign of all UH Mānoa parking policy or transportation infrastructure. Out of scope are:

- Construction of multiple garages across campus
- Elimination of all surface parking campus-wide
- Major campus road reconstruction beyond immediate access adjustments
- Changes to tuition, enrollment, or housing policy
- Comprehensive permit pricing reform for the whole university
- Transit system redesign or new shuttle networks outside temporary mitigation support
- Unrelated academic building construction or interaction with surrounding buildings

This scope statement protects the project from becoming too broad. It keeps the deliverable focused on one specific, realistic site intervention.

HIGH-LEVEL COST SUMMARY

The high-level project cost summary breaks down the one-off costs involved in developing the project, and its recurring costs in operation once the parking structure is completed. The total cost estimate for the one-off aspect of the project comes out to be about \$13 million to \$18 million and covers everything from planning and feasibility study to design and engineering, permits, site preparation, construction, parking technology systems, safety systems, accessibility features, electric vehicle (EV)



charging stations, and operations. These are the major costs involved in building the project and bringing it online.

Since the garage will have to remain in operation even after completion, the recurring costs of the project need to be considered separately from the initial budget. Such costs are related to maintenance, utilities, cleaning, lights, security, CCTV system, EV charging stations, software licenses, sensor maintenance, inspections, and parking management. Distinguishing these two sets of costs highlights the fact that the project is not just construction but also requires continued commitment towards maintaining its operations on campus. This categorization of one-off and recurring costs is shown in [Appendix F](#).

7. TANGIBLE DELIVERABLES

For this project, the tangible deliverables are:

1. A completed multi-level parking structure on the current Shidler–George Hall–Architecture lot
2. A net increase in central-core parking capacity
3. A smart parking occupancy detection system
4. Entrance signage displaying available space counts
5. Permit-controlled access and parking operations integration
6. ADA-compliant accessibility features, including designated stalls and vertical circulation
7. At least 2 EV charging spaces
8. Lighting, CCTV coverage, and emergency communication points
9. Final operations manual, maintenance guidance, and training materials
10. Handoff to the operational department responsible

These deliverables are tangible, measurable, and directly tied to the project goals. The definition of done for the project should have been achieved when the construction of the parking building was completed and had undergone all inspections and approvals before occupancy. At that time, all systems will have been installed and tested and then transferred to the operational responsibility of the proper departments of the business, along with training manuals, maintenance guides, and procedures.

8. REQUIREMENTS ANALYSIS



The UH Mānoa Central Campus Parking Redevelopment project requirements matrix provides a well-rounded and thoughtful approach to modernizing campus parking infrastructure. Spanning 26 total requirements across categories such as Capacity, Accessibility, Safety, Smart Parking, Operations, Sustainability, Compliance, Performance, Project Management, Construction, Security, User Experience, and Schedule, the matrix does well in balancing immediate functional needs with long-term operational goals. Most requirements are classified as "Must," which signals a high-priority, non-negotiable baseline for the project, while a smaller subset is labeled "Should," indicating aspirational but flexible additions such as EV charging spaces and CCTV coverage. Ownership is distributed across Facilities, Parking Services, Campus Security, IT, and the Project Team, all to ensure that no single department carries the full weight of implementation. Furthermore, the inclusion of smart parking features such as real-time occupancy displays, mobile-friendly access, and 99 percent uptime performance standards reflects a forward-thinking design philosophy that goes beyond simply replacing a surface lot. In conclusion, this requirements matrix provides a strong and organized foundation for guiding the project from feasibility through closeout, offering clear traceability between stakeholder needs and actionable deliverables. See [Appendix A](#) for the spreadsheet.

REQUIREMENTS TRACEABILITY

The UH Mānoa Central Campus Parking Redevelopment project traceability matrix provides a clear and organized overview of how each project objective connects directly to its supporting requirements. Spanning five primary project objectives across areas such as parking capacity, smart technology integration, operational efficiency, sustainability, and regulatory compliance, and project and construction management, the matrix does well in demonstrating that no objective exists in isolation from the broader requirement set. Each objective traces back to multiple requirements, with traceability focus areas ranging from capacity, access, and space utilization to design, testing, deployment, and closeout activities, all to ensure that every phase of the project is accounted for from feasibility through construction. Additionally, the coverage check at the bottom of the matrix reinforces accountability by confirming that every requirement from REQ-001 through REQ-026 is linked to at least one project objective, which signals a thorough and well-governed planning process. The primary phases represented across the matrix include Feasibility, Design and Build, Test and Deploy, Promote and Build, and All Phases, reflecting the full project lifecycle. In conclusion, this traceability matrix serves as a strong connective document that bridges high-level project goals with granular technical and operational requirements, providing the project team with a reliable tool for managing scope and ensuring nothing falls through the cracks during execution. See more about this in [Appendix B](#).



9. USER STORIES, USE CASE, AND SWIMLANE DIAGRAMS

USER STORIES

- As a commuter student, I want to know whether spaces are available before I enter the garage so that I do not waste time circling for parking.
- As a faculty member, I want reliable parking near the academic core so that I can arrive at classes and meetings efficiently.
- As a parking administrator, I want real-time occupancy information so that I can manage access and facility use more effectively.
- As a visitor, I want clear entry instructions and wayfinding so that I can park near my destination without confusion.

USE CASE: PARKING IN THE NEW STRUCTURE

Actor: Commuter student

System: UH Mānoa Central Campus Parking Structure

Preconditions: User has a valid permit or authorized parking access. The garage is operational.

Basic Flow:

1. The driver approaches the garage entrance.
2. Digital signs display available spaces.
3. The driver enters the facility.
4. System records facility occupancy update.
5. The driver follows wayfinding to the available level or zone.
6. Driver parks and exits via pedestrian route.
7. System updates occupancy for operational monitoring.

Alternative Flow:

8. The garage is at capacity.



9. The entrance sign shows zero spaces available.
10. Users are redirected to alternate parking guidance or another facility.

Postcondition: User has either parked successfully or been rerouted without needing to circulate through a full facility.

SWIMLANE DIAGRAMS

As part of the development of the project plan, three workflow diagrams in swim lane format were developed to illustrate how major project control processes will interact between stakeholders, the project manager, technical owners, contractors, IT professionals, sponsors, and the Change Control Board. Such workflow diagrams are crucial for our project because we are dealing with building construction, smart parking technology implementation, campus stakeholders, budgetary control, and the transfer of responsibilities. Without workflow diagrams, the process can be chaotic.

Swimlane notation aids in the definition of ownership, indicating who will take care of what part of the process. This is particularly applicable in the case of a large campus infrastructure project because not all decisions will be made by the same individual. There will be some that could be taken care of by the project manager/assigned owner, while there may be others that need to go before a sponsor or even the Change Control Board.

The Project Issue Management Workflow ([Appendix C1](#)) provides details about identifying, recording, assigning ownership, resolving, validating, and closing issues in projects. In other words, this workflow applies more to the build, test, deploy, and closeout phases, as issues can arise in these processes during construction, implementation of the smart parking system, integration of permits, safety testing, or communicating with stakeholders. The first step in this workflow involves a stakeholder or a project team member recognizing an issue. The second step requires the project manager to record, evaluate the severity and urgency, assign the issue, and decide on whether it is feasible to solve it within the current constraints.

The Risk Management Workflow outlined in [Appendix C2](#) illustrates the steps of risk identification, risk assessment, assignment of risk ownership, monitoring, and closure. It directly addresses the project's risk management section, as there are many significant risks involved in the redevelopment of the parking facility, such as issues related to structure/site feasibility, cost overrun, risk of losing parking temporarily while building the structure, traffic flows, stakeholder opposition, technology integration issues, and others. The risks need to be entered in the risk register and reviewed

The Change Management Workflow presented in [Appendix C3](#) depicts how project change requests are submitted, assessed, authorized, implemented, and documented. This is significant since changes might occur within a project involving parking structure development due to requirements for changes regarding drawings, project scope, budget, timeline, smart parking system, accessibility requirements, traffic management, security systems, and construction activities. The process of change initiation involves the identification of an area where a change is required by a stakeholder or member of the project team. This is followed by logging of the request and assessment of impacts on project scope, schedule, budget, risk, and quality. In case of changes falling within current levels of authorization and baseline tolerances, implementation is allowed, but where such limits are exceeded, escalation to the sponsor or the Change Control Board is required.

10. RISK ASSESSMENT AND RISK MITIGATION

The exact footprint may have engineering, drainage, or geotechnical constraints that reduce achievable capacity.

RISK 2: BUDGET OVERRUN RISK

Structured parking is expensive, and the cost per space can rise depending on design complexity and local construction conditions. VTPI's 2025 analysis notes that structured parking is significantly more



expensive than surface parking and becomes more rational where land value is high, and surface parking is inefficient.

Mitigation: Use conceptual cost estimating early, include contingency, and conduct value engineering during design.

RISK 3: TEMPORARY PARKING LOSS DURING CONSTRUCTION

The project site currently provides some parking. Once construction starts, those spaces disappear until the structure is completed.

Mitigation: Create a temporary parking displacement plan, including overflow guidance and phased communication to users.

RISK 4: TRAFFIC FLOW RISK

Adding a garage can increase concentrated turning movements and vehicle queues at access points.

Mitigation: Model ingress and egress in design, improve signage, and adjust circulation controls if needed.

RISK 5: STAKEHOLDER RESISTANCE

Some users may dislike the disruption, cost, or visual impact of a garage.

Mitigation: Use communication management, renderings, capacity comparisons, and evidence-based justification.

RISK 6: TECHNOLOGY INTEGRATION RISK

If sensors, signage, and permit systems do not work together reliably, the smart parking system will not deliver its intended value.

Mitigation: Build a formal test strategy and execute system validation before deployment. Smart parking literature consistently emphasizes integration reliability as a major implementation issue.

RISK 7: NATIVE POPULATION CULTURAL ASPECTS



There is a major risk of a cultural conflict if the existing parking structure is removed and more is built on top. If any investigation of the land leads to any historical findings, the project will be immediately shut down.

Mitigation: There is already a parking lot structure existing, which insinuates that there is permitting and a better chance of completing the project. Projects that are built on already developed land have a much higher chance of getting started and not being cancelled.

11. WORK BREAKDOWN STRUCTURE

The UH Mānoa Central Campus Parking Redevelopment project work breakdown structure provides a comprehensive and logically sequenced roadmap for delivering the project from initial planning through final closeout. Spanning six primary phases, including Feasibility, Design, Build, Test, Deploy, and Close, the WBS does well in organizing the full scope of work into manageable and traceable deliverables that align with the project lifecycle. The Feasibility phase establishes the foundation of the project through five key work packages. F1 creates the Business Needs Assessment by reviewing campus parking conditions, defining the parking access problem, and documenting project justification. F2 creates the Issues and Opportunity Assessment by identifying commuter pain points, analyzing inefficient land use of the current lot, and defining the project opportunity. F3 conducts the Return on Investment analysis by estimating cost per space, estimating net capacity gain, and assessing operational value. F4 conducts Market Research by reviewing the best parking practices, smart parking research, and benchmarking occupancy and cost guidance. F5 conducts the Scope Analysis by defining in-scope and out-of-scope work and establishing project boundaries, all to ensure the project is justified and well-defined before any resources are committed.

The Design phase builds on that foundation through four structured work packages. D1 conducts the Stakeholder Analysis by identifying the needs and interests of students, faculty, staff, Parking Services, and Facilities and Administration. D2 conducts the Budget Analysis by structuring cost estimates for construction, smart systems, and contingency planning. D3 creates the Milestone Map and Timeline by documenting planning, design, and delivery dates. D4 conducts Requirements Elicitation by capturing functional, safety, technology, and accessibility requirements, reflecting a deliberate effort to align project scope with stakeholder needs before construction begins. The Build phase is the most expansive section of the WBS, covering five work packages that balance physical construction with project oversight. B1 executes the Task List through site preparation, foundations, structural framing, deck construction, elevators and stairs, and lighting and systems installation. B2 manages Stakeholder



Engagement and Maintenance through construction updates, user communication, and temporary parking mitigation. B3 creates the Test Strategy and Planning Document by developing structural, smart system, and safety inspection plans. B4 conducts Communication Management and Monitoring through contractor coordination, progress reporting, and issue escalation. B5 conducts Risk Management and Monitoring by tracking budget, schedule, and construction risks, all to ensure execution and oversight are treated as equally important responsibilities throughout the building.

The Test phase introduces a structured quality control process through three work packages. T1 executes the Test Plan through structural testing, lighting and safety testing, occupancy system testing, and permit integration testing. T2 monitors Defect and Issue Management by logging defects, correcting issues, and retesting systems. T3 creates and updates Workflows, Policies, and Procedures covering parking operations, maintenance, and user guidance, reflecting the project team's commitment to delivering a safe and fully operational facility. The Deploy phase transitions the project from construction to active use through three work packages. DP1 releases the product and service by opening the garage, activating the smart parking system, and beginning monitored operations. DP2 finalizes the Training Document by preparing parking staff, security staff, and maintenance personnel for day-to-day operations. DP3 executes the Monitoring and Support Process by tracking occupancy performance, monitoring issues, and supporting users as the facility moves into its operational phase.

In conclusion, the Close phase wraps up the project through three final deliverables. C1 finalizes all Workflows, Policies, and Procedures to ensure operational continuity. C2 transfers the project to the Operational Department, completing the formal handoff from the project team. C3 conducts the Lessons Learned review, providing a clean and accountable ending that captures knowledge gained and positions the facility for long-term success. Please view [Appendix D](#) for figures.

12. TIMELINE / MILESTONE MAP

The UH Mānoa Central Campus Parking Redevelopment project schedule summary provides a clear and well-structured overview of how the project is organized across a 30-month timeline spanning seven major phases, with the longest single-phase clocking in at 13 months. The schedule does well in presenting a logical and sequential flow of work that mirrors the full project lifecycle from initial planning through final closeout. The Feasibility phase covers months 1 through 3, represented by milestones M1 through M3, and concludes with the key handoff of feasibility approval at M3, which signals that the project has received the green light to move forward into more detailed planning. The Design phase follows from months 4 through 8, represented by milestones M4 through M8, and wraps up



with the design package ready handoff at M8, reflecting the completion of all stakeholder analysis, budget planning, milestone mapping, and requirements elicitation activities. The Permits phase occupies months 9 through 11, represented by milestones M9 through M11, and concludes with the permits cleared handoff at M11, all to ensure all regulatory and compliance approvals are secured before any ground is broken.

The Construction phase is the longest and most resource-intensive portion of the schedule, spanning months 12 through 24 across milestones M12 through M24, and concludes with the construction complete handoff at M24. Additionally, the Testing phase follows immediately after construction from months 25 through 26, represented by milestones M25 and M26, and concludes with the accepted handoff at M26, reflecting the project team's commitment to verifying structural, safety, and smart system performance before opening. The Deployment phase covers months 27 through 28, represented by milestones M27 and M28, and concludes with the deployment complete handoff at M28, marking the official activation of the facility and its smart parking systems. In conclusion, the Closeout phase brings the project to a formal end from months 29 through 30, represented by milestones M29 and M30, with the closeout complete handoff at M30 serving as the final accountability checkpoint that transfers the project to operations and documents lessons learned for future reference.

13. NEXT STEPS

If this idea were accepted for further examination, the next step would involve taking the concept developed during the planning phase and turning it into something that can be formally reviewed and accepted by the university. The first step should involve presenting the concept to all the right people at the university, which would include Facilities, Parking Services, campus administration, and anyone else who might be affected by the project. In this way, the university could determine whether this site, purpose, cost, and benefit were worth exploring further.

After initial review, the university should begin a formal feasibility and site investigation. This would include a detailed site survey, geotechnical review, drainage review, traffic circulation analysis, and early engineering assessment to confirm whether the selected lot can realistically support a multi-level parking structure. At the same time, the project team should refine the budget estimate, evaluate funding options, and begin a more detailed stakeholder communication plan. This is especially important because the project would temporarily remove existing parking during construction and could create concerns related to cost, disruption, traffic flow, and campus appearance.

Once feasibility is confirmed, the project should move into design and permitting. During this phase, the university would develop concept drawings, confirm the number of levels and parking spaces, define smart parking system requirements, and begin the required permit and approval process. The final pre-construction step would be to create a temporary parking displacement plan, finalize the construction schedule, and prepare communication materials for students, faculty, staff, visitors, and nearby



departments. These next steps would help turn the proposal from a project management concept into an actionable campus infrastructure project.

14. CONCLUSION

The proposed redevelopment of the surface parking lot between Shidler College of Business, George Hall, and the Architecture Building into a multi-level parking structure is a strong project candidate from both an operational and project management perspective. The evidence does not suggest that UH Mānoa has no parking whatsoever. Instead, it shows something more useful and more defensible: campus parking operates under meaningful peak-hour pressure, upper-campus areas have historically approached full utilization during working hours, and permit sellouts indicate constrained access for student drivers.

This specific lot is an attractive site because it already functions as parking and sits inside the academic core, where proximity matters most. Replacing it with a structured facility would improve land-use efficiency, increase core-campus parking capacity, and reduce parking uncertainty for commuters. When paired with smart parking technology, the project would also improve the facility's usability by providing real-time information and clearer operations.

Most importantly, this project aligns well with the logic of the University of Hawaii at Manoa Campus. It supports a complete needs assessment, a scope analysis, a detailed and testable requirements framework, a risk assessment, a predictive WBS, and a milestone-based timeline. In that sense, it is not only a good infrastructure idea. It is also a strong project management proposal.

APPENDICES

APPENDIX A: REQUIREMENTS ANALYSIS

UH MĀNOA CENTRAL CAMPUS PARKING REDEVELOPMENT

Requirements Matrix | REQ-001 to REQ-013 | Date Added: March 2026

Total Requirements: 26	Must Have: 21	Should Have: 5	Phase Coverage: Feasibility through Closeout
------------------------	---------------	----------------	--

ID	Category / Function	Priority	Owner	Requirement Statement
REQ-001	Capacity	Must	Facilities	Replace the existing surface parking lot with a multi-level parking structure on the designated site.
REQ-002	Capacity	Must	Parking Services	Provide a minimum of 300 total parking spaces.
REQ-003	Capacity	Must	Parking Services	Provide a net gain of at least 200 spaces above the pre-project site condition.
REQ-004	Accessibility	Must	Facilities	Include ADA-compliant parking spaces in accordance with applicable code.
REQ-005	Accessibility	Must	Facilities	Include at least one elevator serving all parking levels.
REQ-006	Accessibility	Must	Facilities	Include stair access to all levels and clearly marked pedestrian routes to surrounding academic buildings.
REQ-007	Safety	Must	Campus Security	Provide lighting coverage for all parking bays, vehicle circulation paths, and pedestrian routes.
REQ-008	Safety	Must	Campus Security	Provide CCTV coverage at entry points, stairwells, elevator lobbies, and parking levels.
REQ-009	Safety	Should	Campus Security	Include emergency call or assistance points in public user areas.
REQ-010	Smart Parking	Must	IT / Parking Services	Detect and report parking-space occupancy in real time.
REQ-011	Smart Parking	Must	IT / Parking Services	Update occupancy status within 10 seconds of a vehicle entering or exiting a monitored space or facility zone.
REQ-012	Smart Parking	Must	Parking Services	Display available-space counts at the garage entrance.
REQ-013	Smart Parking	Should	IT	Allow users to view facility availability through a mobile-friendly website or application.

REQ-014	Operations	Must	Parking Services	Integrate with existing UH permit and access procedures.
REQ-015	Operations	Must	Parking Services	Include directional signage for entry, exit, level guidance, and pedestrian egress.
REQ-016	Operations	Must	Parking Services	Include a documented operations and maintenance workflow prior to turnover.
REQ-017	Sustainability	Must	Facilities	Include energy-efficient lighting.
REQ-018	Sustainability	Should	Facilities / Sustainability	Include EV charging spaces.
REQ-019	Compliance	Must	Facilities	Comply with applicable Hawaii building, fire, accessibility, and life-safety requirements.
REQ-020	Performance	Must	IT	Maintain at least 99 percent uptime for the parking information system during normal operating periods.
REQ-021	Project Mgmt.	Must	Project Team	Complete feasibility, design, build, test, deploy, and close phases in accordance with the predictive WBS structure used in class.
REQ-022	Project Mgmt.	Must	Project Team	Produce stakeholder analysis, budget analysis, milestone map, risk log, test plan, and lessons learned documentation.
REQ-023	Construction	Must	Project Team	Include a temporary parking-displacement mitigation plan before demolition or closure of the existing lot.
REQ-024	Security	Should	Campus Security	Retain recorded security footage for an operationally defined period approved by the responsible department.
REQ-025	User Experience	Must	Parking Services	Include clear wayfinding from garage exits to adjacent academic buildings.
REQ-026	Schedule	Must	Project Team	Complete the project within 30 months from project initiation.



APPENDIX B: REQUIREMENTS TRACEABILITY

UH Manoa Central Campus Parking Redevelopment Project

Two-Page Requirements Traceability Matrix (RTM) | Objective-to-Requirement Coverage

6 Project Objectives	26 Requirements Mapped	REQ-001-026 Coverage Range	100% Traceability Coverage
-------------------------	---------------------------	-------------------------------	-------------------------------

Page 1: Objective-to-Requirement Coverage Summary

Project Objective	Requirement IDs	Traceability Focus	Primary Phase(s)	Count
Increase parking capacity on central campus	REQ-001, REQ-002, REQ-003	Capacity, access, and space utilization	Feasibility / Design	3
Ensure accessibility and safety of the parking facility	REQ-004, REQ-005, REQ-006, REQ-007, REQ-008, REQ-009	Accessible access, safety controls, circulation, and facility security	Design / Build / Test	6
Implement smart parking technology and reliable system performance	REQ-010, REQ-011, REQ-012, REQ-013, REQ-020	Smart parking tools, permit integration, uptime, and data accuracy	Design / Build / Test / Deploy	5
Support efficient operations, navigation, and user experience	REQ-014, REQ-015, REQ-016, REQ-025	Wayfinding, operations procedures, user guidance, and support readiness	Design / Deploy	4
Ensure sustainability and regulatory compliance	REQ-017, REQ-018, REQ-019	Sustainability, environmental requirements, permitting, and compliance	Design / Permits / Build	3
Ensure proper project management and controlled construction impacts	REQ-021, REQ-022, REQ-023, REQ-024, REQ-026	Budget, schedule, communications, risk control, handoff, and closeout	All Phases	5
Coverage Check Every requirement ID from REQ-001 through REQ-026 is mapped to a project objective.	Management Use Use this matrix to verify that scope, design, testing, deployment, and closeout work trace back to project goals.	Change Control Any added, removed, or changed requirement should be reflected in this matrix before approval.		

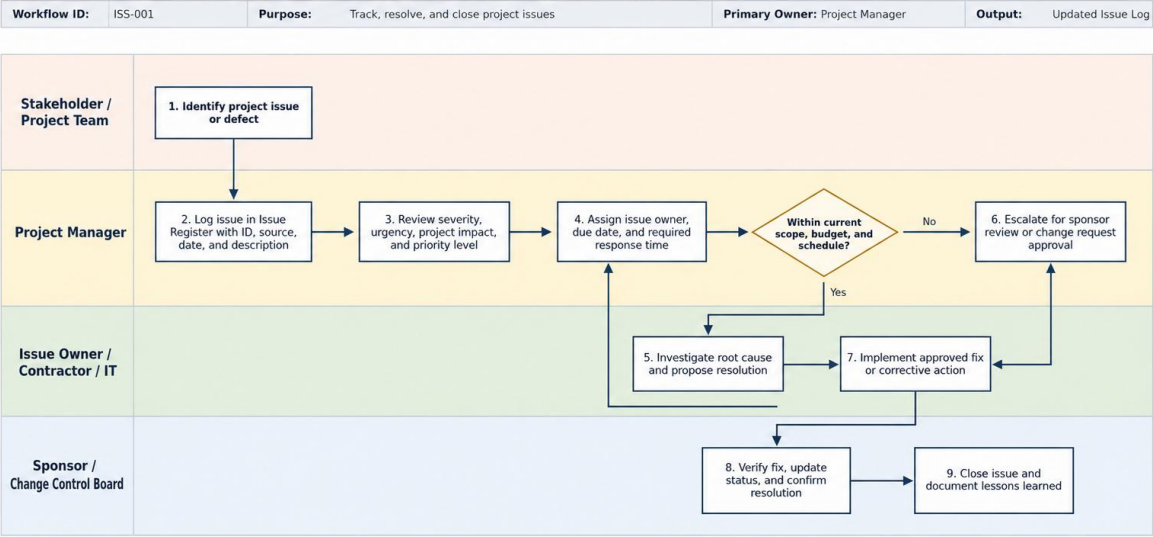


APPENDIX C: SWIMLANE DIAGRAMS

APPENDIX C1: ISSUE MANAGEMENT WORKFLOW

UH MĀNOA CENTRAL CAMPUS PARKING REDEVELOPMENT

Swimlane Workflow | Issue Management | Date Added: May 2026 | Phase Coverage: Build through Closeout



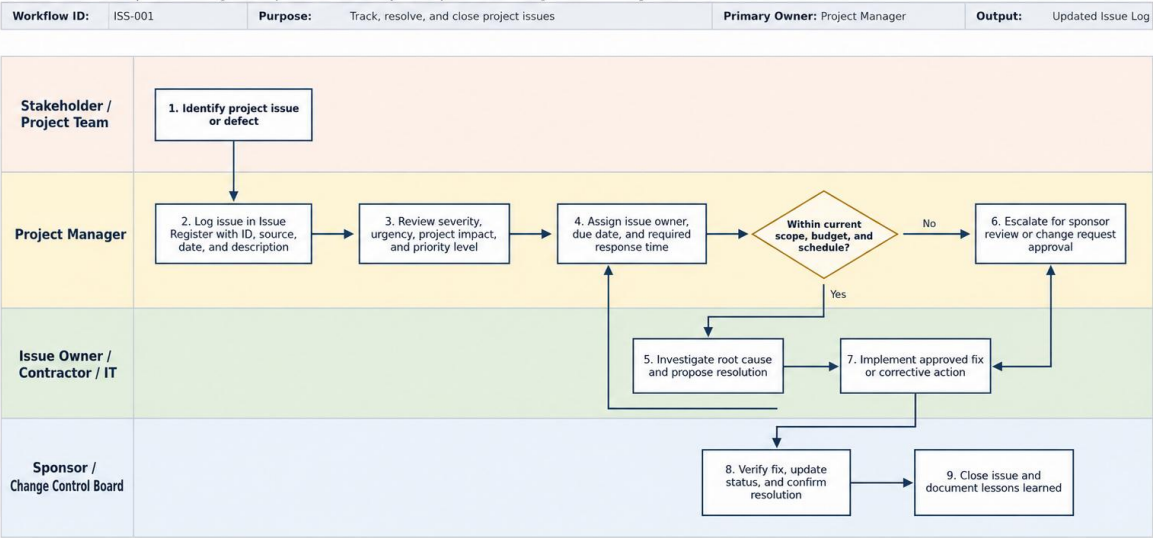
Closure criteria: The issue may be closed only after the fix is verified, the issue register is updated, and any required communication is sent to affected stakeholders.



APPENDIX C2: RISK MANAGEMENT WORKFLOW

UH MĀNOA CENTRAL CAMPUS PARKING REDEVELOPMENT

Swimlane Workflow | Issue Management | Date Added: May 2026 | Phase Coverage: Build through Closeout



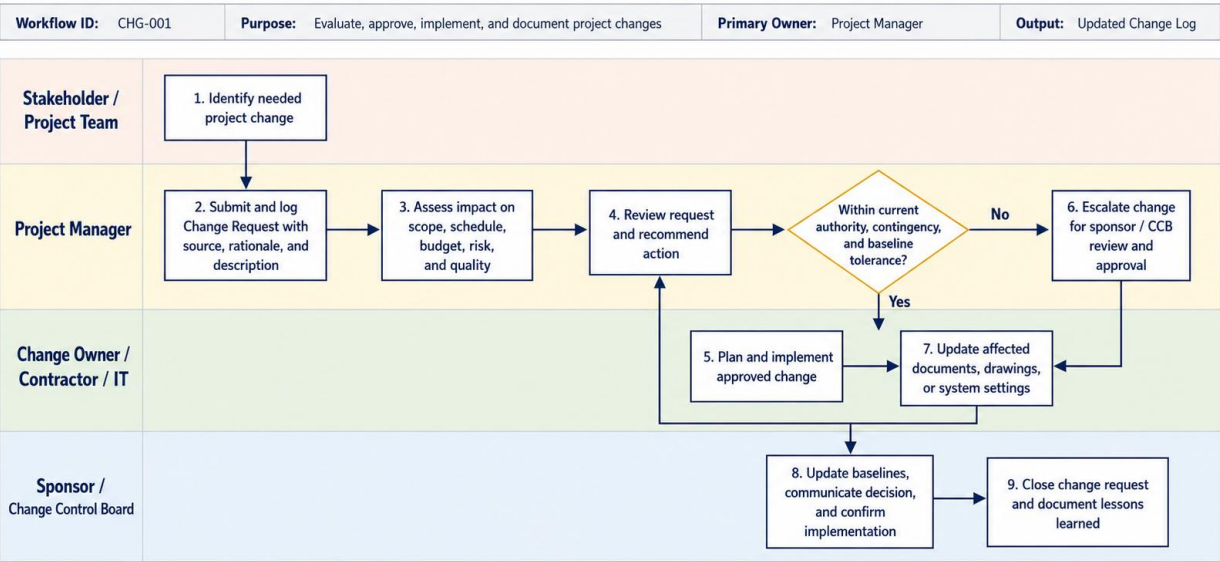
Closure criteria: The issue may be closed only after the fix is verified, the issue register is updated, and any required communication is sent to affected stakeholders.



APPENDIX C3: CHANGE MANAGEMENT WORKFLOW

UH MĀNOA CENTRAL CAMPUS PARKING REDEVELOPMENT

Swimlane Workflow | Change Management | Date Added: May 2026 | Phase Coverage: Build through Closeout



Closure criteria: A change request may be closed only after the decision is documented, approved changes are implemented, affected project records are updated, and required stakeholder communication is completed.



APPENDIX D: WORK BREAKDOWN STRUCTURE

UH Manoa Central Campus Parking Redevelopment Project

Two-Page Work Breakdown Structure (WBS) | Feasibility - Design - Build - Test - Deploy - Close

6 Phases	23 Work Packages	76 Key Activities	1.0 Draft Version
-------------	---------------------	----------------------	----------------------

Purpose: organize the parking redevelopment effort into clear phases, work packages, and deliverables for scope control, project tracking, construction coordination, testing, launch, and handoff.

Page 1: Planning and Early Build Work Packages

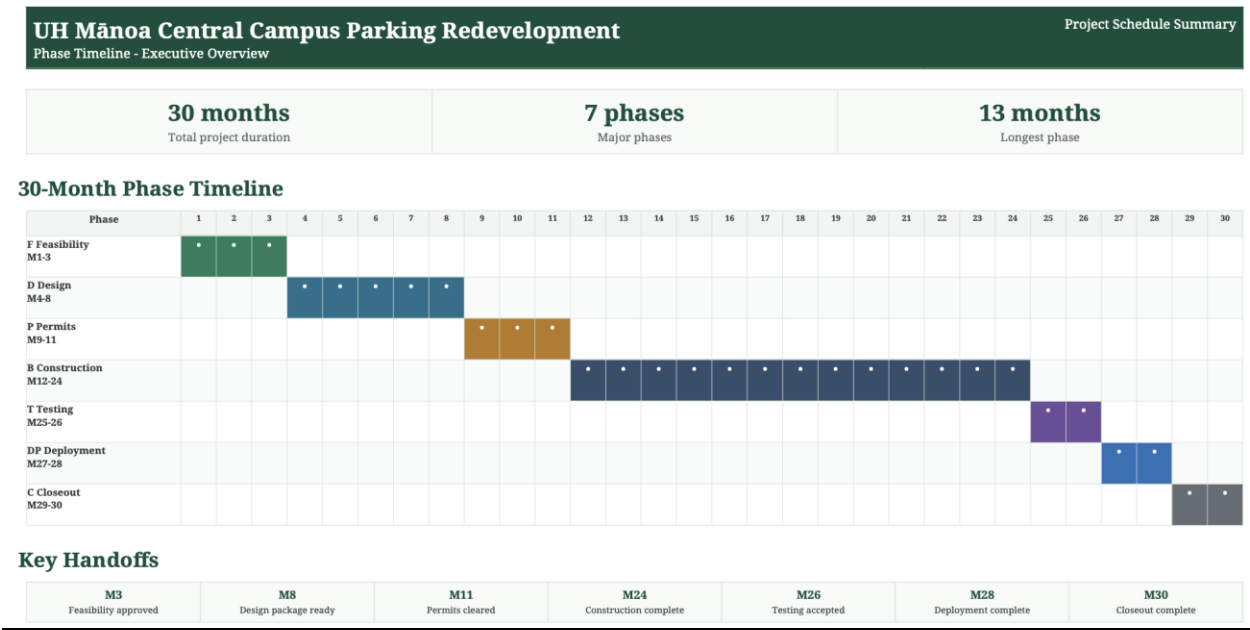
WBS	Work Package	Key Activities / Deliverables
F. Feasibility - Validate the need, opportunity, value, and project boundaries.		
F1	Business Needs Assessment	Review campus parking conditions; Define parking access problem; Document project justification
F2	Issues / Opportunity Assessment	Identify commuter pain points; Analyze inefficient current land use; Define redevelopment opportunity
F3	Return on Investment Analysis	Estimate cost per space; Estimate net capacity gain; Assess operational value
F4	Market Research	Review parking best practices; Review smart parking research; Benchmark occupancy and cost guidance
F5	Scope Analysis	Define in-scope work; Define out-of-scope work; Establish project boundaries
D. Design - Translate the approved opportunity into requirements, budget, and timeline documentation.		
D1	Stakeholder Analysis	Students; Faculty; Staff; Parking Services; Facilities / Administration
D2	Budget Analysis	Structure cost estimate; Estimate smart system costs; Estimate contingency reserves
D3	Milestone Map and Timeline	Planning milestones; Design milestones; Delivery milestones
D4	Requirements Elicitation	Functional requirements; Safety requirements; Technology requirements; Accessibility requirements
B. Build - Execute construction, coordination, communication, and risk management activities.		
B1	Construction Task Execution	Site preparation; Foundation work; Structural frame construction; Deck construction; Elevators and stairs; Lighting and system installation
B2	Stakeholder Engagement	Construction updates; User communication; Temporary parking mitigation

Page 2: Build Completion, Testing, Deployment, and Closeout

WBS	Work Package	Key Activities / Deliverables
B3	Test Strategy and Planning	Structural inspection plan; Smart system test plan; Safety inspection plan
B4	Communication Management	Contractor coordination; Progress reporting; Issue escalation
B5	Risk Management and Monitoring	Budget monitoring; Schedule monitoring; Construction risk tracking
T. Test - Verify the structure, systems, safety elements, and operating procedures before launch.		
T1	Test Plan Execution	Structural testing; Lighting and safety testing; Occupancy system testing; Permit integration testing
T2	Defect and Issue Management	Log defects; Correct issues; Retest systems after correction
T3	Workflow, Policy, and Procedure Updates	Parking operations procedure; Maintenance procedure; User guidance procedure
DP. Deploy - Launch the garage, train staff, and stabilize early operations.		
DP1	Release Product / Service	Open garage; Activate smart parking system; Begin monitored operations
DP2	Training Materials	Parking staff training; Security staff training; Maintenance staff training
DP3	Monitoring and Support	Track occupancy performance; Monitor operational issues; Support users during launch
C. Close - Finalize handoff, operating documentation, and lessons learned.		
C1	Finalize Workflows, Policies, and Procedures	Confirm final operating procedures; Archive approved process documentation; Publish final user guidance
C2	Transfer to Operational Department	Complete ownership handoff; Transfer support responsibilities; Confirm operational acceptance
C3	Lessons Learned	Review project successes and challenges; Document improvement opportunities; Create future project recommendations



APPENDIX E: TIMELINE



UH Mānoa Central Campus Parking Redevelopment

Project Schedule Summary

Phase Register - Focus, Outputs, and Closeout Points

Phase Register

Phase	Timeframe	Primary Focus	Key Outputs
F Feasibility	Months 1-3	Confirm need, opportunity, value, and scope boundaries.	Needs assessment; issue/opportunity assessment; ROI analysis; market research; scope analysis.
D Design	Months 4-8	Translate the approved concept into stakeholder, budget, schedule, and requirements documentation.	Stakeholder analysis; budget analysis; milestone map; functional, safety, technology, and accessibility requirements.
P Permits	Months 9-11	Obtain required approvals before physical redevelopment work begins.	Permit applications; code review responses; approval tracking; final permit authorization.
B Construction	Months 12-24	Build the garage structure and install required building, safety, and smart-parking systems.	Site preparation; foundations; structural frame; deck construction; elevators/stairs; lighting and systems.
T Testing	Months 25-26	Verify the structure, lighting, occupancy technology, safety controls, and procedures.	Structural testing; safety testing; occupancy-system testing; defect logs; retesting and acceptance.
DP Deployment	Months 27-28	Open the service, activate smart systems, train support teams, and begin monitored operations.	Garage opening; smart-parking activation; staff training; user support and occupancy monitoring.
C Closeout	Months 29-30	Complete the transition to operations and document lessons learned.	Final workflows; operations handoff; closeout documentation; lessons learned report.

Schedule Management Notes

1. Use Months 1-11 to reduce uncertainty before construction begins.	2. Treat Month 24 as a major acceptance gate before testing and deployment.	3. Closeout should confirm ownership transfer, procedures, support model, and lessons learned.
--	---	--



APPENDIX F: COST BREAKDOWN

UH Mānoa Central Campus Parking Redevelopment			Project Cost Summary
High-Level Cost Summary - One-Time Project Costs and Ongoing Operational Costs			
High-Level Cost Register			
Cost Area	One-Time Project Cost Drivers	Planning Estimate	Ongoing Operational Cost Drivers
Planning & Feasibility	Site surveys, geotechnical review, circulation analysis, feasibility study, early ROI estimate.	\$150K–\$350K	Not a recurring cost after approval.
Design, Engineering & Permits	Architectural design, structural engineering, code review, permitting, accessibility review.	\$1.2M–\$2.5M	Future redesigns only if scope changes.
Site Prep & Construction	Demolition/clearing, utilities, foundations, 4-level structure, ramps, stairs/elevators, deck work.	\$13M–\$20M	Long-term structural repairs included under maintenance.
Smart Parking Technology	Occupancy sensors, digital availability signage, permit-system integration, testing, activation.	\$600K–\$1.5M	\$60K–\$150K/year software, sensor support, replacements.
Safety, Accessibility & Sustainability	Lighting, CCTV, emergency call stations, ADA parking/vertical circulation, at least 2 EV spaces.	\$500K–\$1.2M	\$75K–\$200K/year security, inspections, EV/lighting upkeep.
Operations Transition	Operations manual, staff training, handoff package, deployment support, closeout documentation.	\$100K–\$250K	\$150K–\$350K/year staffing/admin support if assigned.
Contingency / Owner Reserve	Initial acceptance reserve and owner contingency for project uncertainty.	\$1.5M–\$3.0M contingency	\$200K–\$450K/year maintenance; \$80K–\$180K/year utilities/cleaning.
TOTAL (One-Time Project Cost)		\$13M–\$18M	See ongoing annual costs at right.



APPENDIX G: PROJECT ROLES

Category	Role	Key Responsibility	Organizational Owner
University Leadership	VP of Administration & Finance	Ultimate project sponsor and budget authority; provides final approval for capital expenditure and signs off on major project decisions	University of Hawai'i at Mānoa
University Leadership	Director of Facilities & Capital Improvements	Day-to-day owner representative; manages consultant and contractor relationships on behalf of the university and ensures project aligns with campus standards	UH Facilities
University Leadership	Director of Commuter Services / Parking	End-user lead and operational requirements owner; defines parking operations needs, permit integration requirements, and coordinates user communication during construction	UH Commuter Services
Design Team Leads	Principal Architect / Design Principal	Head of the architecture firm leading the project; responsible for all architectural design, permit drawings, code compliance, and coordination with engineering subconsultants	Architecture Firm (TBD via RFP)
Design Team Leads	Principal-in-Charge - Structural Engineering	Senior structural engineer of record; leads seismic, wind load, and gravity design for the 4-level structure in accordance with Hawaii building codes	Structural Engineering Firm (TBD)
Design Team Leads	Principal Civil Engineer	Heads site and civil scope including grading, drainage, utility connections, and access point design adjacent to the Shidler-George Hall-Architecture corridor	Civil Engineering Firm (TBD)
Construction Leadership	Construction Executive / Senior Project Manager (GC)	Head of the General Contractor's team under a Construction Manager at Risk delivery model; oversees scheduling, subcontractor management, safety compliance, and budget performance during the Build phase	General Contractor (TBD via RFP)
Construction Leadership	Project Executive - Owner's Representative	Acts as the owner's primary field advocate during design and construction; manages quality, schedule, and contract compliance on behalf of the university	Owner's Rep Firm (if engaged)
Regulatory & Compliance	State DAGS Project Administrator	Required state oversight contact for public construction on university-owned land; engagement must begin during the Permits phase (months 9-11)	State of Hawaii - DAGS
Regulatory & Compliance	Honolulu DPP Plan Review Chief	Municipal permitting authority; approves building permit applications and construction documents for the new multi-level structure	City & County of Honolulu
Finance & Procurement	UH Chief Financial Officer	Approves capital project funding, oversees bond issuance or state appropriation requests, and ensures the project budget is authorized within university financial governance	University of Hawai'i System
Finance & Procurement	Procurement Officer	Heads the competitive RFP and bidding process for all major contracts including design, construction, and smart parking technology; ensures compliance with public procurement law	UH Procurement Office



REFERENCES

- Federal Highway Administration. (2021, March 29). *Overview of parking and access pricing strategies*. U.S. Department of Transportation.
https://ops.fhwa.dot.gov/publications/fhwahop08044/cp_prim6_04.htm
- Federal Highway Administration. (2020, July 1). *Contemporary approaches to parking pricing: A primer* (Sections 3, 5, 7). U.S. Department of Transportation.
<https://ops.fhwa.dot.gov/publications/fhwahop12026/>
- Biyik, C., Allijn, R., Leib, A., Lim, M., Sheth, N., Washington, E., & Speyer, O. (2021). Smart parking systems: Reviewing the literature, architecture, and ways forward. *Smart Cities*, 4(2), 623–642.
<https://www.mdpi.com/2624-6511/4/2/32>
- Fahim, A., Hasan, M., & Chowdhury, T. (2021). Smart parking systems: Comprehensive review based on various aspects. *Heliyon*, 7(11), e08123.
<https://www.sciencedirect.com/science/article/pii/S2405844021011531>
- Litman, T. (2025). *Comprehensive parking supply, cost and price analysis*. Victoria Transport Policy Institute. <https://www.vtpi.org/pscp.pdf>
- University of Hawai‘i at Mānoa. (2012). *Campus transportation demand management plan*.
<https://manoa.hawaii.edu/planning/TDM2012.pdf>
- University of Hawai‘i at Mānoa. (2024). *2023–2024 Common Data Set*.
https://manoa.hawaii.edu/miro/wp-content/uploads/2024/08/AnalysisBrief_CommonDataSet_2023-2024_updated_no_data_gender.pdf
- University of Hawai‘i at Mānoa Commuter Services. (2022, January 10). *Spring 2022 student permits sold out*. <https://manoa.hawaii.edu/commuter/2022/01/10/spring-2022-student-permits-sold-out/>
- University of Hawai‘i at Mānoa Commuter Services. (2022, July 5). *Fall student 2022 permit information*. <https://manoa.hawaii.edu/commuter/2022/07/05/fall-student-2022-permit-information/>